



COM 682 Cloud Native Development

Module Introduction

Dr Zeeshan Tariq
z.tariq@ulster.ac.uk

Dr Rashid Kamal
r.kamal@ulster.ac.uk



- Consideration of different cohorts in room
- BB Announcements

This is not a pure theoretical module.
Practical and practise is equally important





Return to Campus

Things are going to be a little different as we return to campus. Read more about [what you need to know](#).

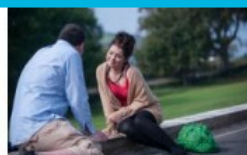
It's ok not to feel ok

Student Wellbeing



Counselling

Find out more about our counselling service and who to call when you are in need.



Triage

Our triage staff will help and guide you to the service you need.



Disability

We provide a variety of services for a range of disabilities to help you make the most of your time at Ulster University.

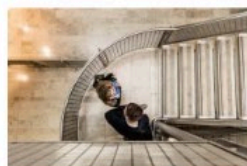
S

We have money.



Health and wellbeing

Find out more on how you can improve your health and wellbeing.



Sexual health clinic

Protect your sexual health by visiting one of our clinics.



Support programmes

View our support programmes.



Staff resources

Resources for University staff.



Student Well Being

<https://www.ulster.ac.uk/wellbeing>

#ReadyForUU

Module Topics and Timetable

Mondays:	Lecture (all students)	15:15 to 17:15
	Lab (PT & 6050)	17:15 to 19:15 (BC-03-311)
Tuesdays:	Lab (SE GrpA)	15:15 to 17:15 (BC-03-303, BC-02-303)
	Lab (CS, SE GrpB)	17:15 to 19:15 (BC-03-311, BC-03-303, BC-03-305, BC-03-309)

Coursework 1 – 50%

Submission Deadline: Week-6 (31st Oct, Friday noon)

Solution Designing:

Design a scalable, cloud native, web-application which acts as a multimedia sharing platform facilitating sharing of multimedia content for **your chosen use case**. The exact media type(s) which can share (from what's covered in the labs) are at the discretion of the student. The ability to share multiple content types is encouraged.

Each student will individually submit a slide deck which details the designed solution (15 slides max)

Coursework 2 – 50%

Submission Deadline: Week-13 (9th Jan 2026, Friday noon)

Implementation of design:

Implement, deploy, and test the **same solution** designed in CW1. This should be implemented and deployed using the Microsoft Azure cloud platform taught and used within the practical exercises associated with this module.

Each student will individually submit 5 mins video of the built solution

- 1) **Written Assignment (50%) submission Week-6**
- 2) **Project (50%) submission week-13**

These coursework assignments are detailed below.

Coursework 1 – Written Assignment [50%]



Coursework 1 – BRIEF

- Students will individually **design** a cloud-native solution in the form of a web app.
- The designed solution should leverage a range of cloud-native technologies and concepts as taught within the module and module materials.

Solution Designing:

Design a scalable, cloud native, web-application which acts as a **multimedia sharing platform** facilitating sharing of multimedia content for **your chosen usecase**. The exact media type(s) which can share (from what's covered in the labs) are at the discretion of the student. The ability to share multiple content types is encouraged.

The **designed** solution of your chosen usecase, should leverage a range of cloud native technologies and concepts as taught within the module and module materials. You may choose to design a **serverless architecture using Azure Functions** as the core compute instead of VMs/App Service.

Your design should include the following, but not limited to:

- Design a multimedia hosting webpage as a Cloud service where the user can upload multimedia files.
- You should choose to create page wireframes. Present this in the form of website design layout diagram. (Covered during week-5 lecture)

Page 10-18
From Module Handbook,
already at BB

Appendix III. Coursework 1 Submission Checklist

Before submitting, make sure you have addressed the following:

General

- Did you include a *title slide* with project name, one-line description of your usecase, your name, and student number?
- Is your submission in *PowerPoint format* and within *15 slides* (excluding references)?

Problem & Context

- Did you *discuss the problem* related to your usecase that you are solving? This should be a hint that why you are using Cloud Computing.
- Did you *identify issues related to scalability* of resources in your project?

Solution Design

- Did you include a *solution architecture diagram* using only *Azure resources*?
- Did you design and present a *multimedia hosting webpage layout* (wireframe or design diagram)?
- Did you include a *network/architecture diagram* showing compute, storage, networking, etc.?
- Did you design a *database schema* (list of resources and/or ERD diagram) for storing user entries?
- Did you design the *REST API* for CRUD operations and present it in a *UML diagram*?
- Did you integrate (or at least propose integration of):
 - Static HTML hosting of a webpage frontend?
 - REST endpoints for service logic and storage connections.
 - Use of both *SQL* and *NoSQL* database structures?

Critical Evaluation

- Did you provide an *overview of advanced features* you intend to develop in the final solution?
- Did you include an *assessment of limitations* of your designed solution?
- Did you include an *assessment of scalability* (horizontal/vertical scaling, autoscaling, etc.)?
- Did you justify *why you selected specific Azure services* over alternatives?

Conclusion & References

- Did you include *concluding comments* summarising your design?
- Did you provide *references* to sources, frameworks, or Azure documentation you used?

☒ If you can tick all these boxes, your submission is likely to meet the expectations. Please submit this checklist at the end of the PowerPoint slide deck. Checklist slide will not be counted in the 15 slides limit.

Please be aware of the declaration about submission of CW:

Plagiarism

N.B. Students should be aware of the plagiarism policy of the University and submit their coursework in accordance with this. Plagiarism is the unattributed copying of the work of another person, either from a published work or the work of another student. It is a form of literary theft and is not permitted under any circumstances. Plagiarism is regarded by the University as a very serious offence and subject to formal disciplinary proceedings. See [5] for more details.

I declare that this is all my own work. Any material I have referred to has been accurately referenced and any contribution of Artificial Intelligence technology has been fully acknowledged. I understand the importance of academic integrity and have read and understood the University's General Regulation: Student Academic Integrity and the Academic Misconduct Procedure. I understand that I must not upload my work before, during or after submission to any unapproved plagiarism detectors or answer sharing platforms, or equivalent, and that only University-approved platforms should be used.

ACCESS YOUR TIMETABLE

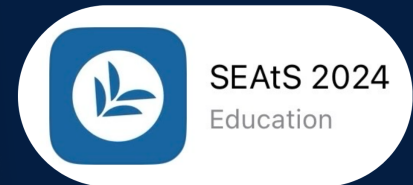


Scan the QR code above to download “Publish”
and access your personalised timetable.

Discover everything you need to know about your
timetable:

ulster.ac.uk/timetabling

DON'T FORGET TO CHECK-IN!



Download “SEAtS 2024” via App Store/Play Store
now and register your class attendance.

Discover everything you need to know about checking
into class:

ulster.ac.uk/attendance

Cloud Native Development

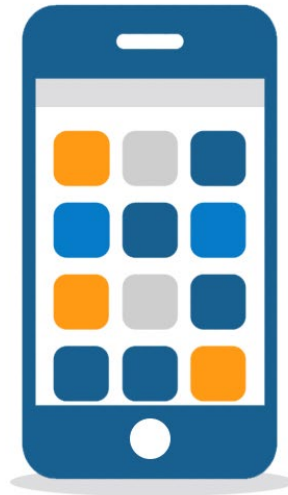
[illegible]

The Purpose of this Module is to explore:

**Modern, Scalable, Cloud based,
Development and deployment practices**

As such, this module has 2 primary areas of focus:

1. Scalable Cloud architecture as a service paradigm
2. Modern development, testing and deployment practices



2.87
MILLION

95% FREE
Google Play

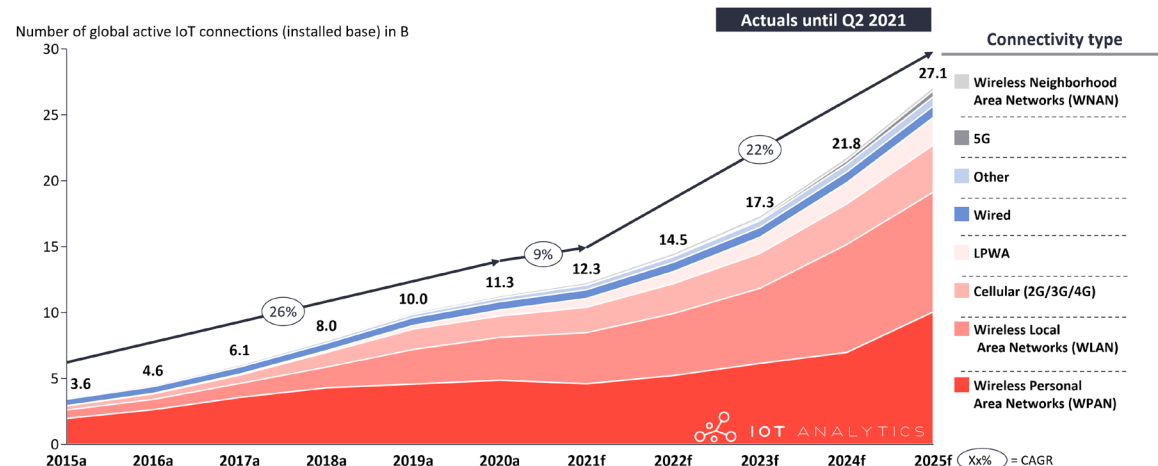


1.96
MILLION

90% FREE
Apple App Store

Module Overview

Global IoT market forecast (in billion connected IoT devices)



Note: IoT Connections do not include any computers, laptops, fixed phones, cellphones or tablets. Counted are active nodes/devices or gateways that concentrate the end-sensors, not every sensor/actuator. Simple one-directional communications technology not considered (e.g., RFID, NFC). Wired includes ethernet and fieldbuses (e.g., connected industrial PLCs or I/O modules). Cellular includes 2G, 3G, and 4G. LPWAN includes unlicensed and licensed low-power networks. WPAN includes Bluetooth, Zigbee, Z-Wave, or similar. WLAN includes Wi-Fi and related protocols. WNAN includes non-short-range mesh, such as Wi-SUN. Other includes satellite and unclassified proprietary networks with any range.

Source: IoT Analytics Research, September 2021 – Please remember to cite IoT Analytics as the source (with link) when re-sharing this content as per our copyright policy

Global business apps
downloads increased
by
90%
on March 15, 2020
compared to their
2019 average



Source: App Annie, 2020

How much you are aware of these technologies/Terms?

- ***Cloud Computing***
- ***Cloud Infrastructure***
- ***Development Operations (DevOps)***
- ***Containerization****
- ***Docker***
- ***Kubernetes***
- ***Microservices***
- ***Serverless Infrastructure***

The UK-US Tech Prosperity Deal

- **Transatlantic AI Pact:** Accelerates world-leading AI research to develop new drugs, faster life-saving treatments, and improved cancer care.
- **Nuclear Technology Alliance:** Marks a golden age of nuclear energy, providing clean, homegrown power and creating high-paying jobs in the UK.
- **North East AI Growth Zone:** Expected to create 5,000+ jobs and attract billions in private investment.
- **Major Industry Deal:** UK firm Nscale partners with NVIDIA and OpenAI to deliver *Stargate UK*.
- **US Tech Investment:** American firms commit over £31 billion to UK AI and tech infrastructure, including Microsoft's largest-ever UK investment.

Nvidia and Microsoft bosses hail huge UK AI investment -

BBC News

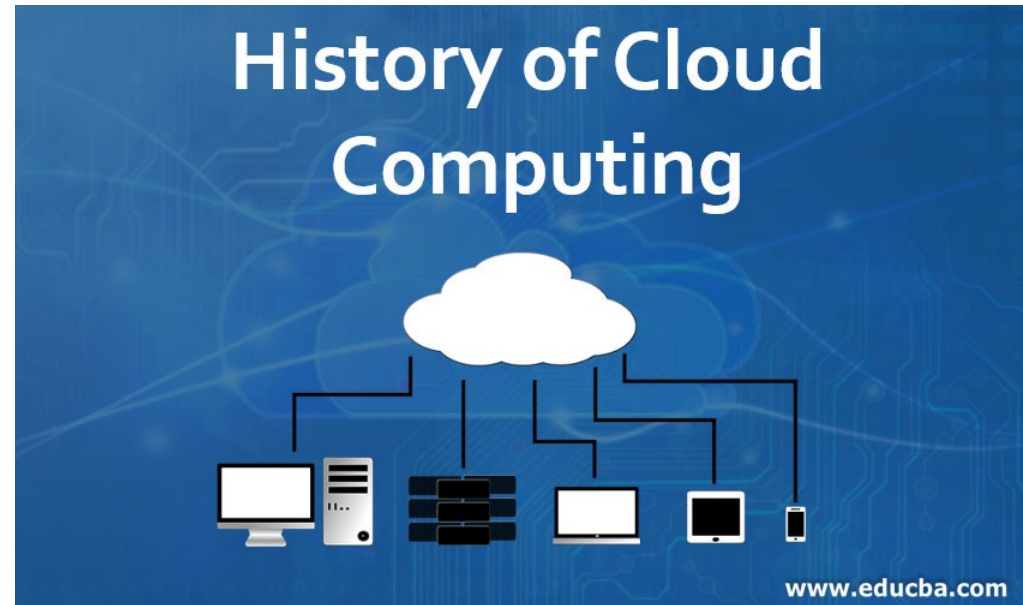
US-UK pact will boost advances in drug discovery, create tens of thousands of jobs and transform lives - GOV.UK



New Data Centre in Derry/LondonDerry

- **New Data Centre in Derry:** Positioned near Ulster University, offering students and researchers opportunities to work with **cutting-edge cloud infrastructure**.
- **Cloud & Connectivity:** Direct link to **Project Kelvin Transatlantic fibre** provides low-latency access to international cloud networks.
- **Renewable-Powered Infrastructure:** 100 MW data center powered by **renewable energy** → supports sustainable cloud-native workloads.
- **Talent & Skills:** Ulster University graduates are highlighted as a key **talent pipeline** for careers in data centers, cloud engineering, and AI.

New Data Centre in Derry to attract 'some of the largest global companies and investors'



An Abridged History of Networked Services and Architecture

In the context of this module, there have been 5 paradigm shifts related to networked services and communications*.

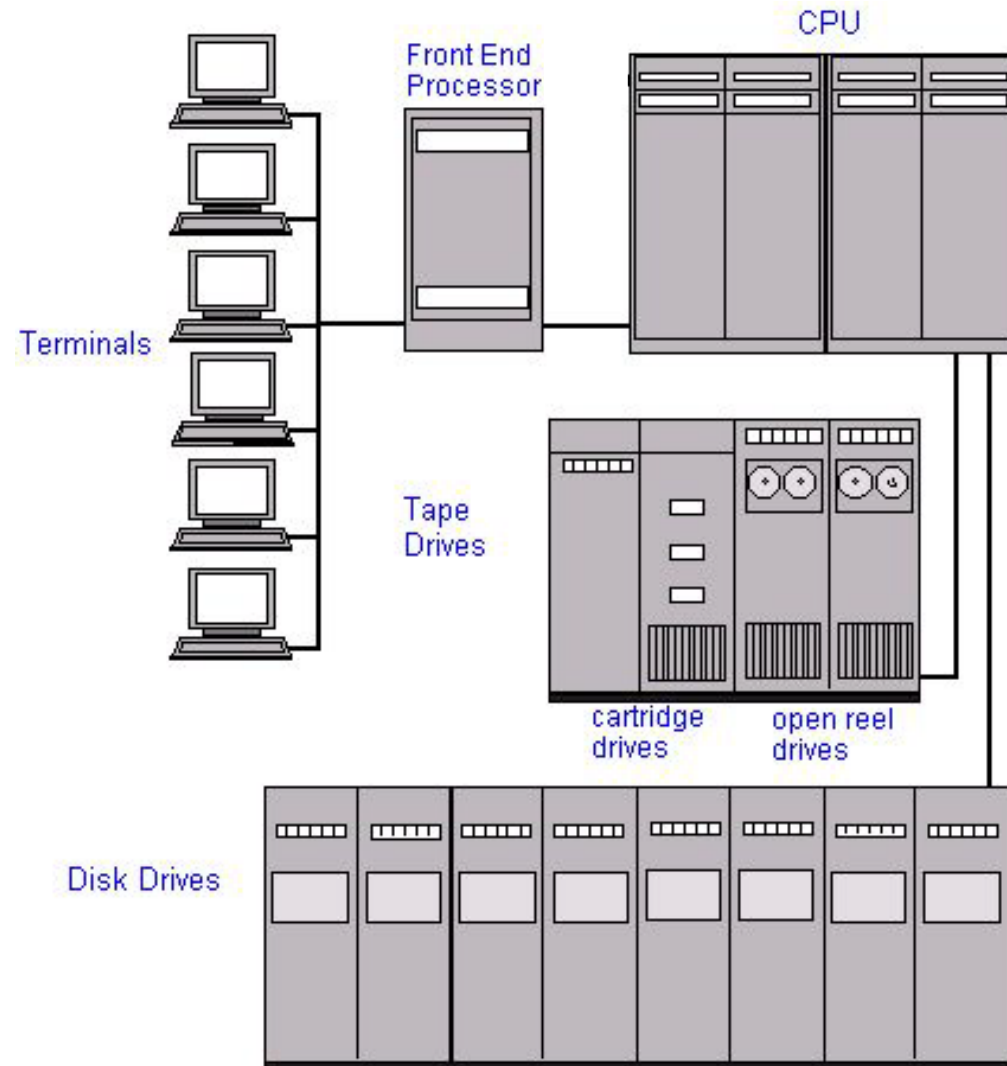
Decade	Emerging Paradigm
1950-1960s	Mainframe and Terminals
1980s	Client-Server
1990s	Web and Hypertext
2000s	Service Oriented Architecture
2010s	Cloud, Fog and Edge Computing

**broadly speaking*

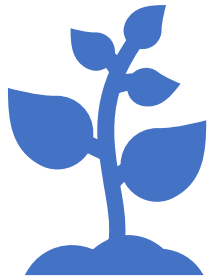
1950s - 1960s
Mainframes and Terminals



Mainframe Architecture



Visionary thinking



1950

Herb Grosch suggested that 15 data centers could service the computing needs for the world



1960

John Mc Carthy proposed that computation may be provided as a public utility, such as water or electricity



1966

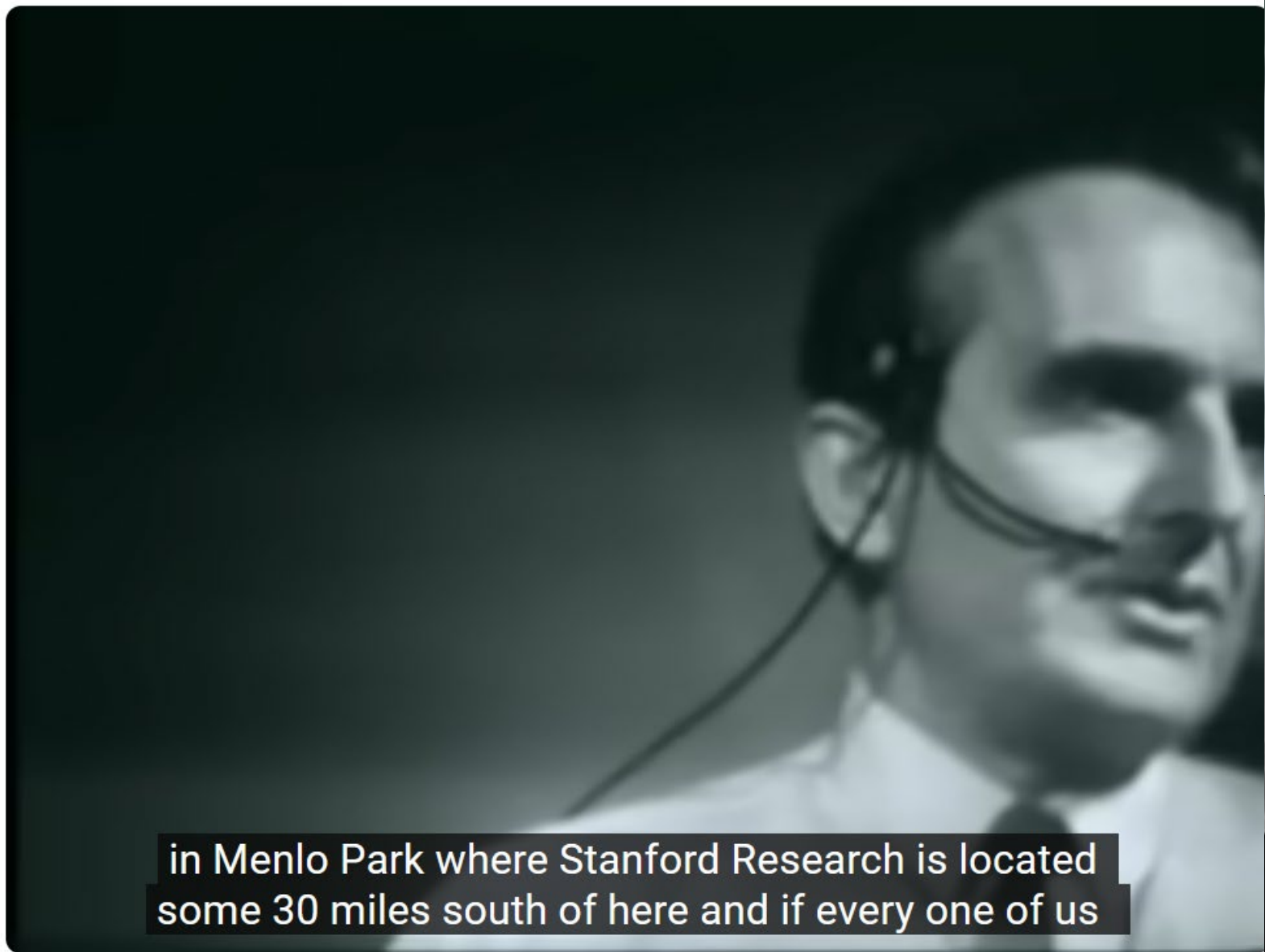
Douglas Parkhill essentially described modern cloud computing in “The challenge of the Computer Utility”

1968

The Mo

by Dougl

vision for the f



in Menlo Park where Stanford Research is located
some 30 miles south of here and if every one of us

The Mother of All Demos, presented by Douglas Engelbart (1968)



Marcel

1.21K subscribers

Subscribe

14K



Share

Save





1969

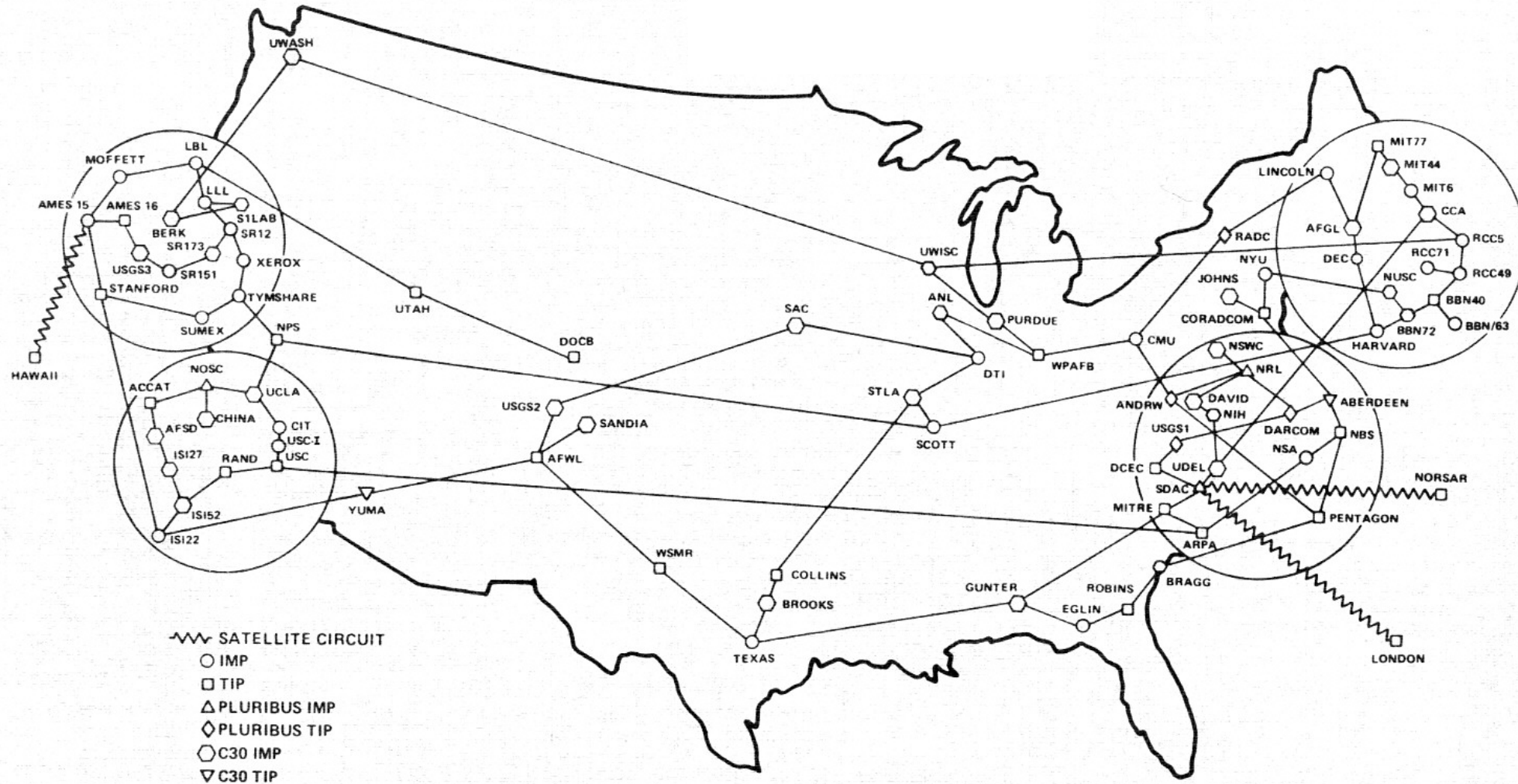
The ARPANET

the predecessor to the internet

packet switching, protocol forwarding

ARPA Network, Geographic Map
September 1973





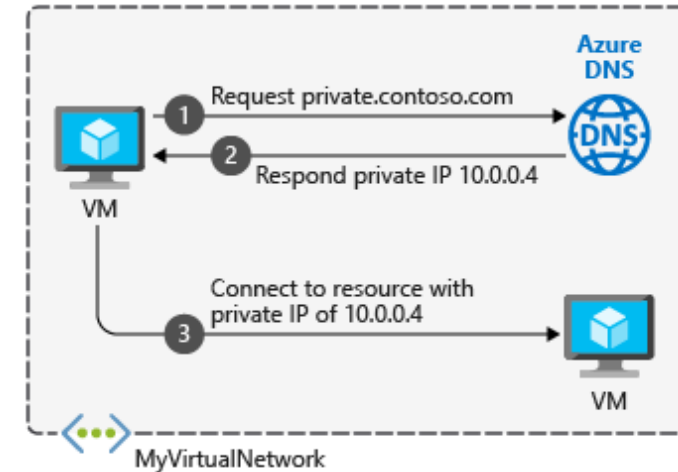
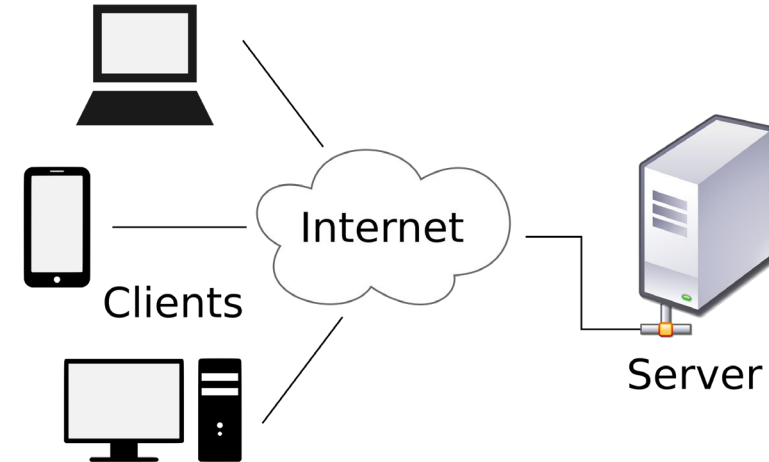
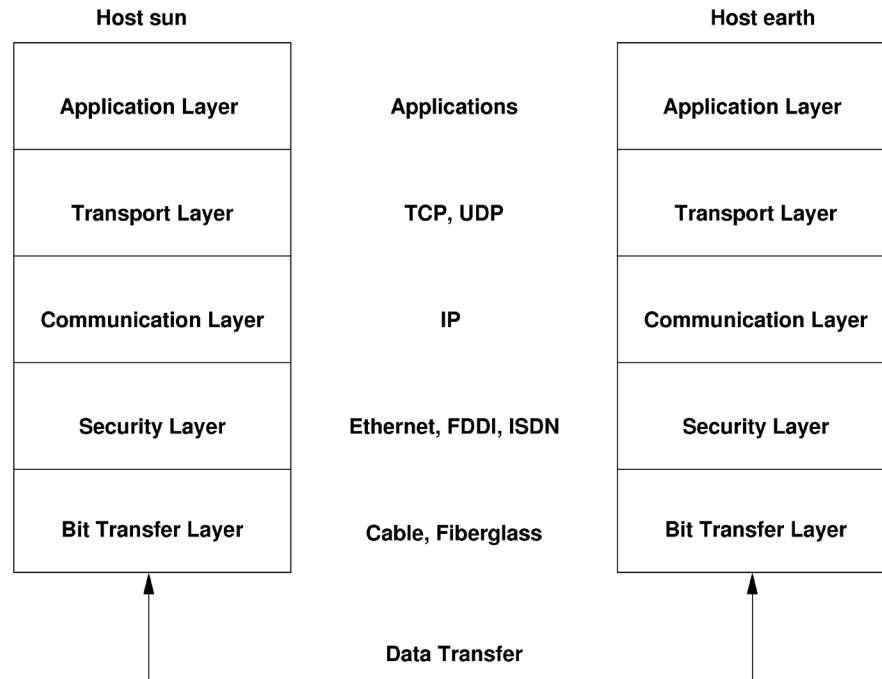
(NOTE: THIS MAP DOES NOT SHOW ARPA'S EXPERIMENTAL SATELLITE CONNECTIONS)
NAMES SHOWN ARE IMP NAMES, NOT (NECESSARILY) HOST NAMES

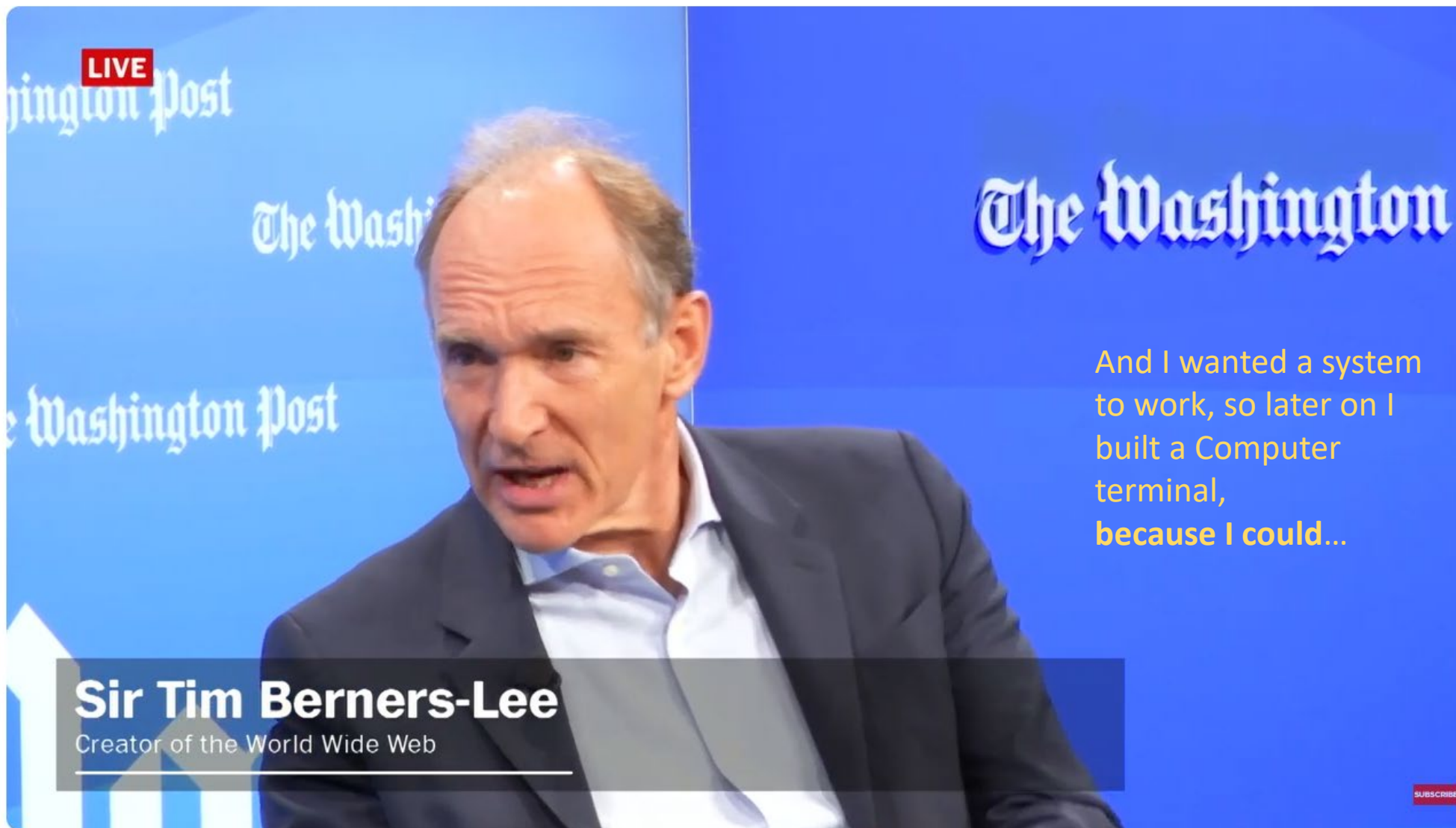
197
TCP



Proc
Aug 2 / Aug 3
LENGTH, DEST
↓
TCP
H
→ PROCESS
TCP
↓
IMP
EST | SE
HOST |
ABS

Client-Server/Internet Building Blocks





Sir Tim Berners-Lee on how he came up with the Internet | Washington Post Live



Washington Post Live ✓
84.7K subscribers

Subscribe

318



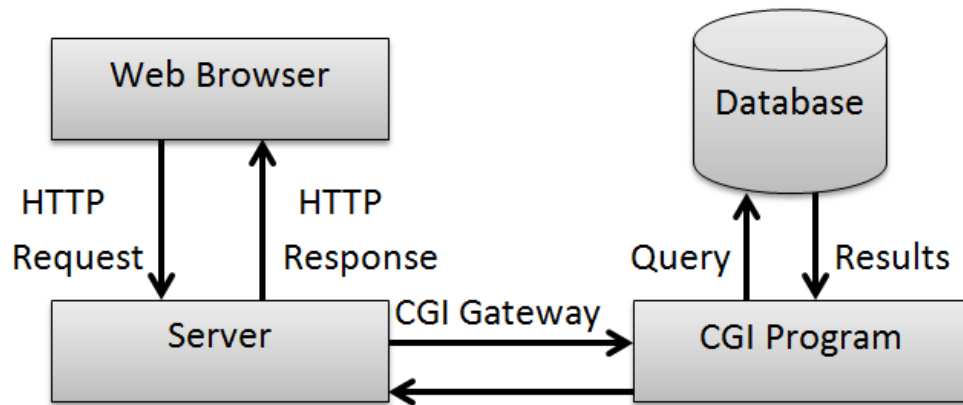
Share

Download

Save



in the



Server-Side Scripting/programming – 1993(ish)

dynamically serve contents within websites

Google emerged and designed software to consume and index the content of the web, automatically.

The first-ever Google server was **built in a custom case made out of Legos**.

The server was housed on the Stanford campus and its web address was initially google.stanford.edu. But Brin and Page registered the Google.com domain name on September 15th, 1997



Google! - 1998

Managing the web

This is how Google looked back in the 90s (Picture: Google)

Google!

Search 1,326,920,000 web pages

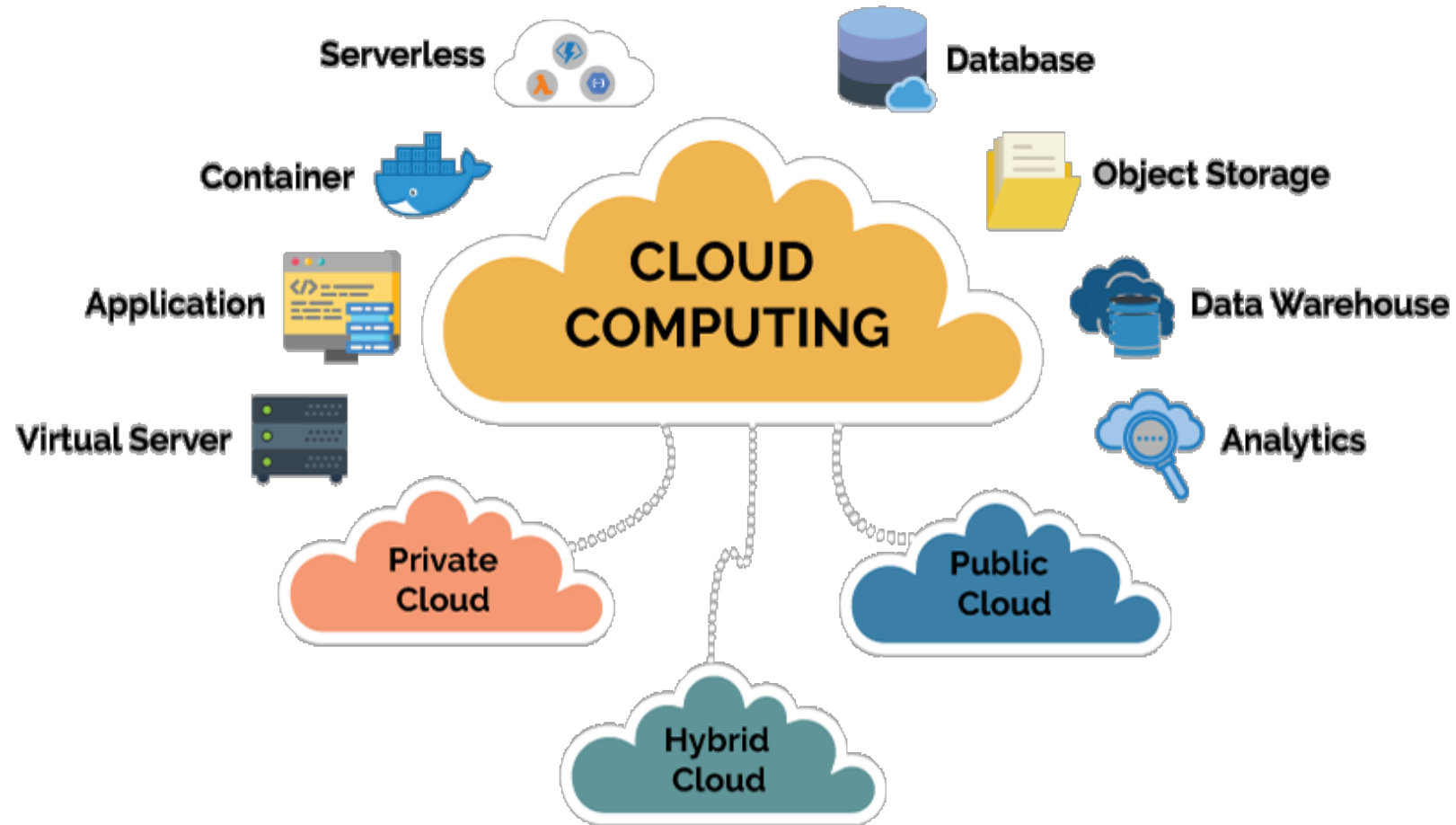
Google Search

2006 Cloud Computing

To overcome Service-oriented architecture overheads, 70% of time spent was IT infrastructure and MGMT

In 2006, one of the largest digital retail companies in the world found it had a need for flexible storage and computation to service its own enterprise. Amazon developed EC2 and S3 to service this. They then opened the AWS





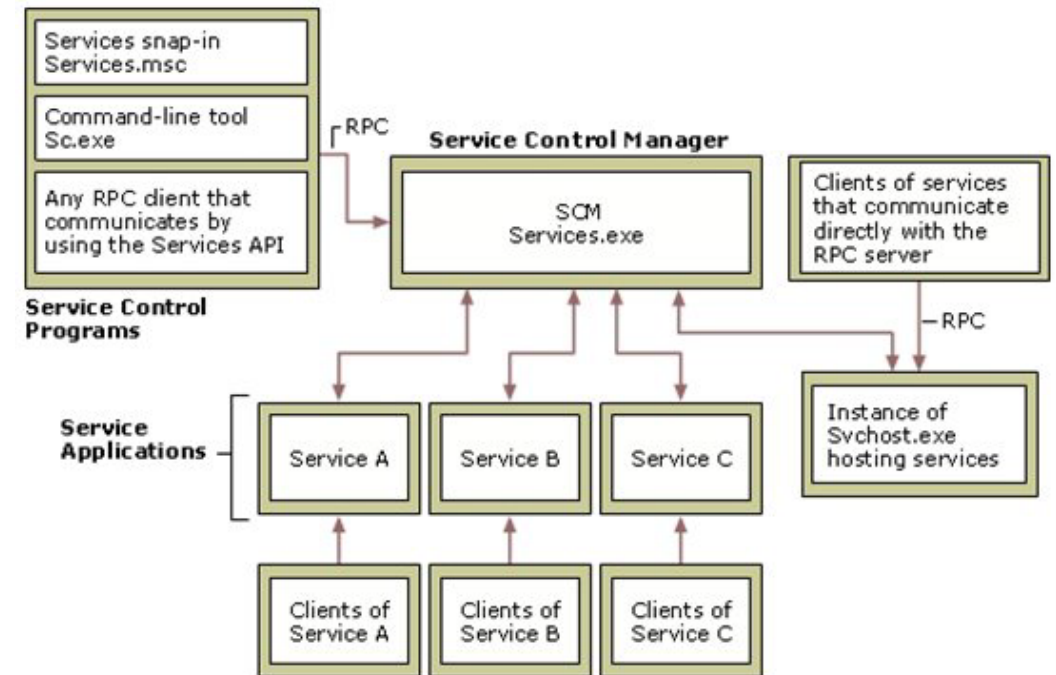
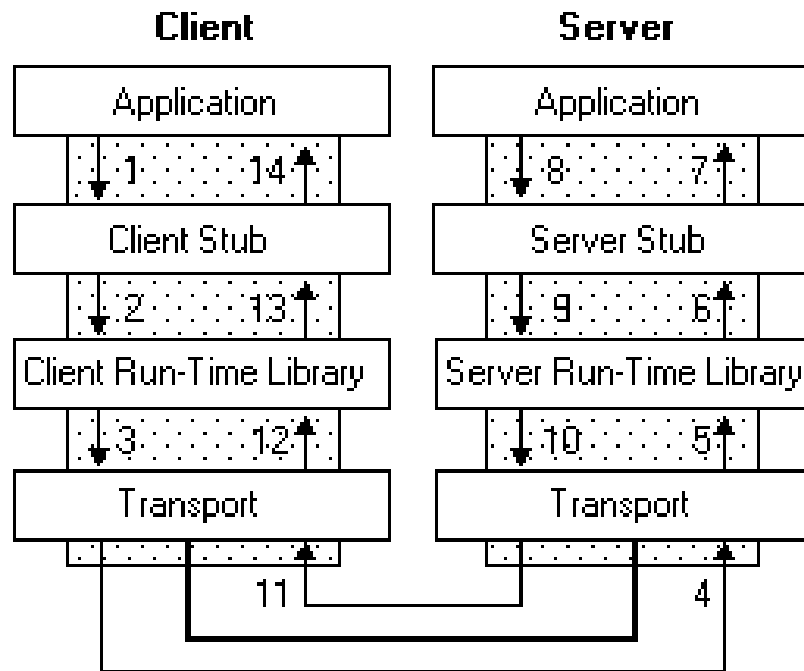
Establishing a common language/protocol

So in addition to the infrastructure to power the storage and computation, which is essentially the core of cloud computing a **communication protocol was needed**.

Historic approaches to networked service protocols

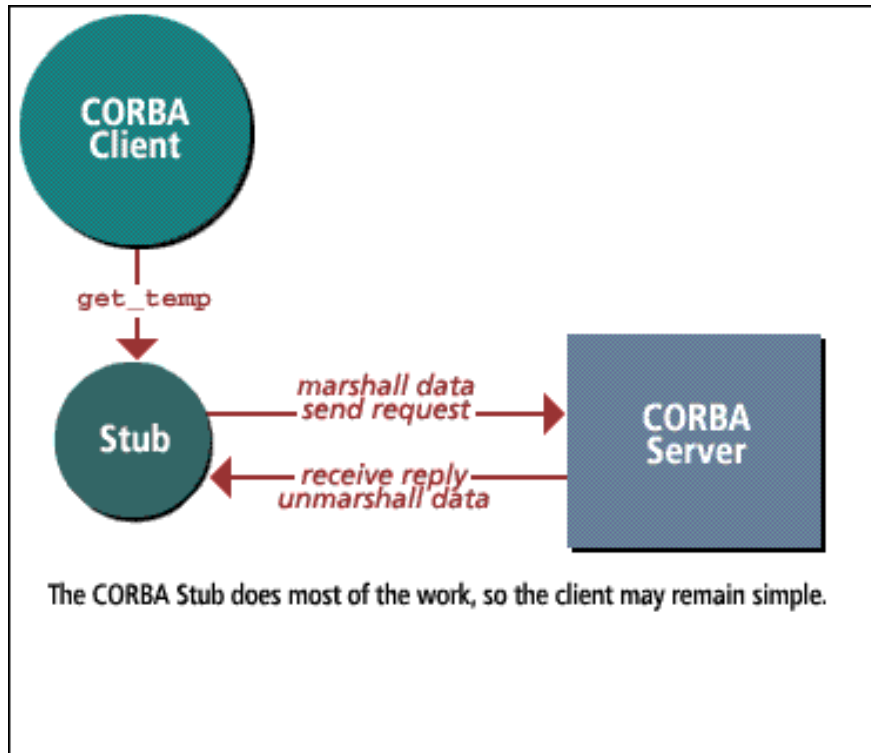
MS Remote Procedure Calls (RPC)

Remote Procedure Call is a software communication protocol that one program can use to request a service from a program located in another computer on a network without having to understand the network's details. RPC is used to call other processes on the remote systems like a local system.



Historic approaches to networked service protocols

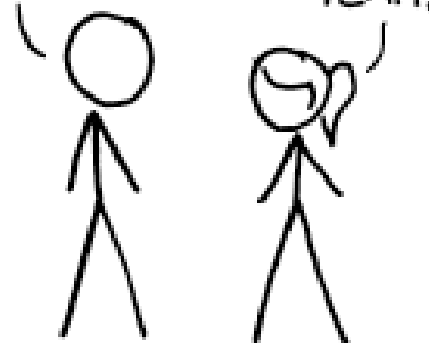
CORBA / Java RMI (Remote Method Invocation)



HOW STANDARDS PROLIFERATE: (SEE: A/C CHARGERS, CHARACTER ENCODINGS, INSTANT MESSAGING, ETC)

SITUATION:
THERE ARE
14 COMPETING
STANDARDS.

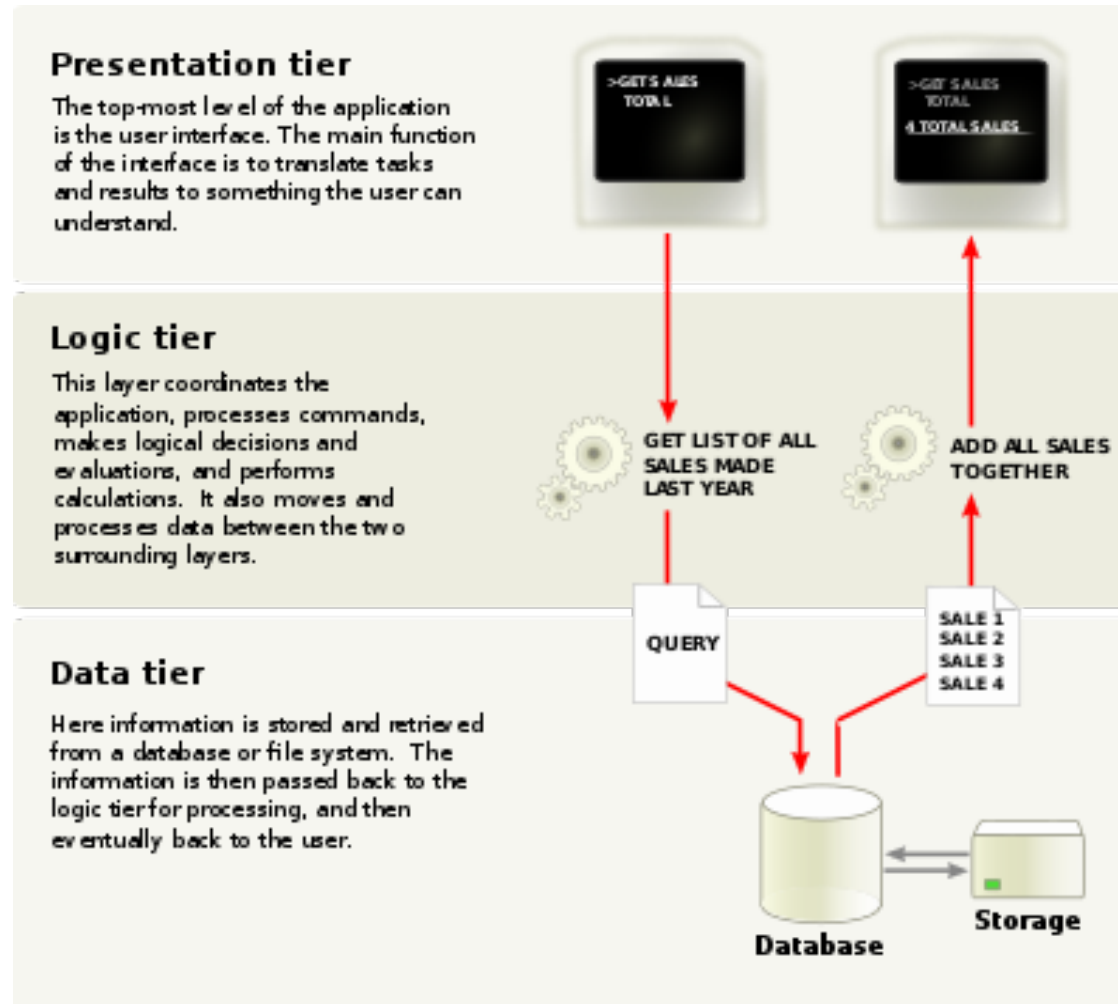
14?! RIDICULOUS!
WE NEED TO DEVELOP
ONE UNIVERSAL STANDARD
THAT COVERS EVERYONE'S
USE CASES.



SOON:

SITUATION:
THERE ARE
15 COMPETING
STANDARDS.

3-Tier Architecture

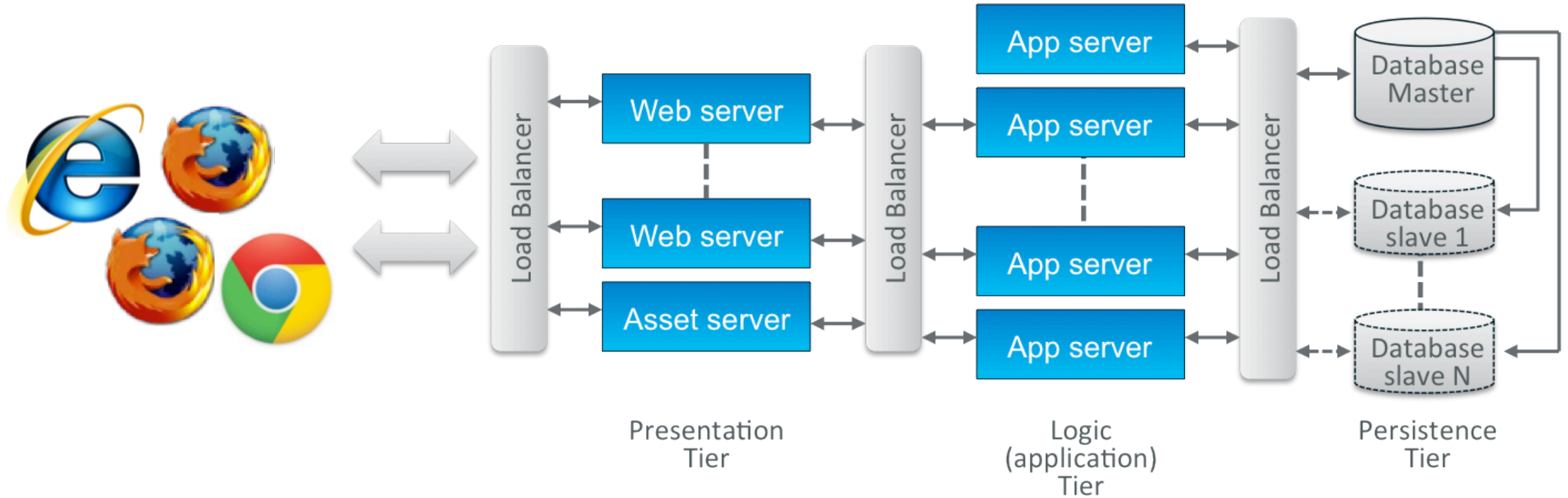


Web Browser
HTML/CSS/JS

Application Server
Node.js, Apache/PHP,
IIS/.net

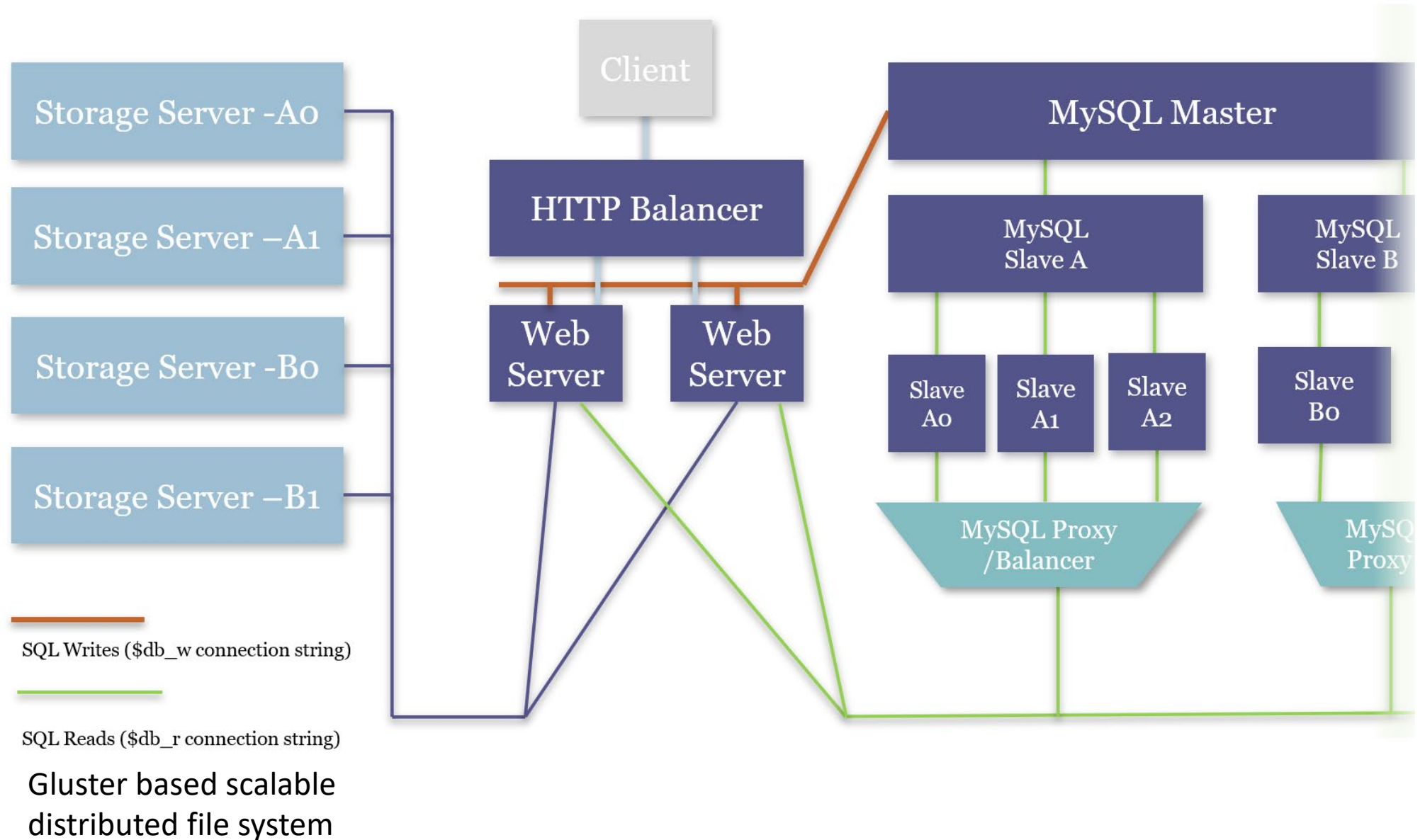
Database/Filesystem
MySQL/Oracle/Mongo
Local filesystem/ceph

Scaling a 3-Tier Architecture

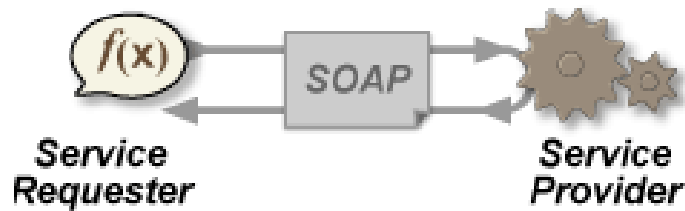


Challenges: Not a sustainable performance oriented solution

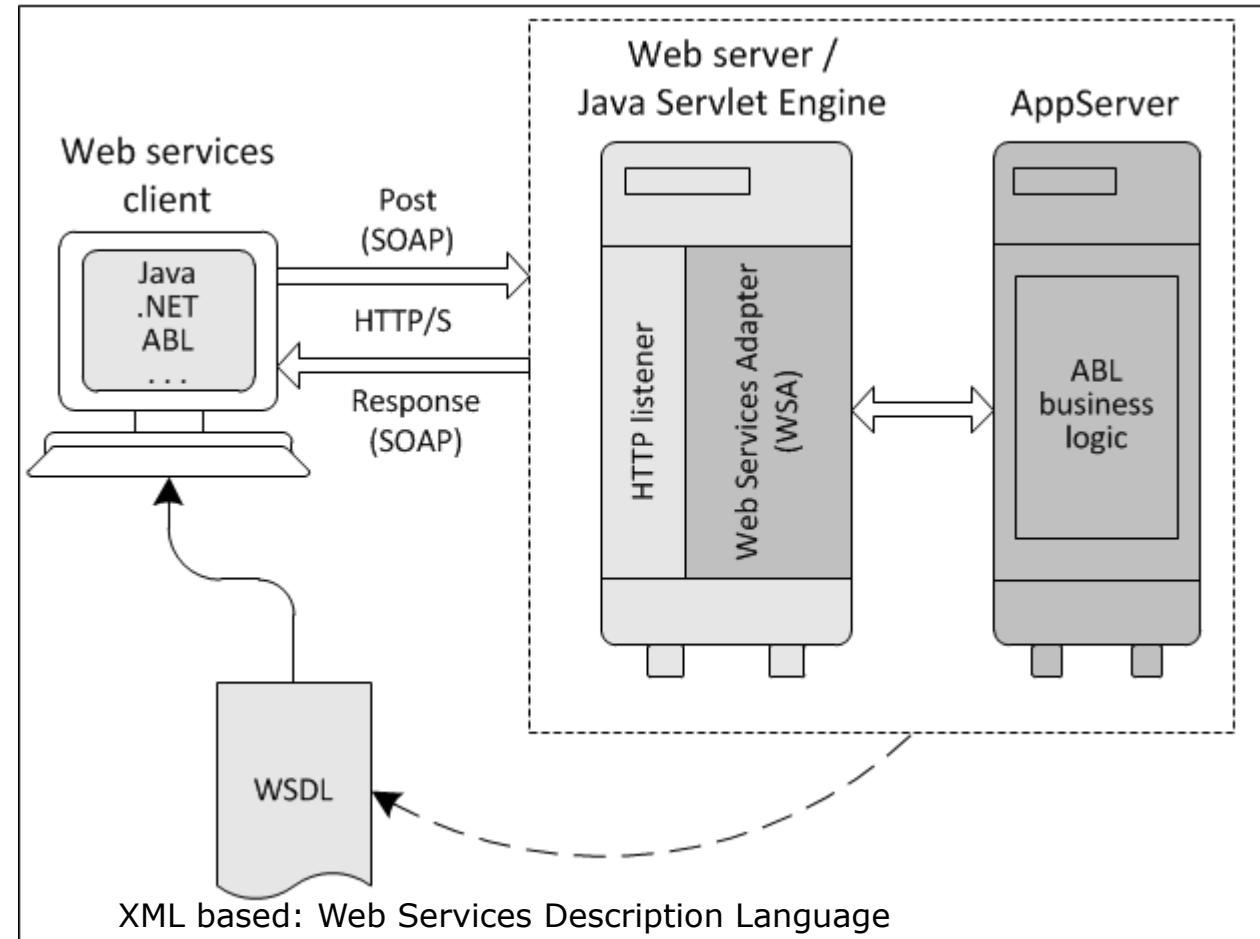
(Slightly) Advanced version of 3-Tier Architecture



Platform Agnostic (generic) Web APIs



SOAP is one of the original platform agnostic, web based, API solutions for web services.

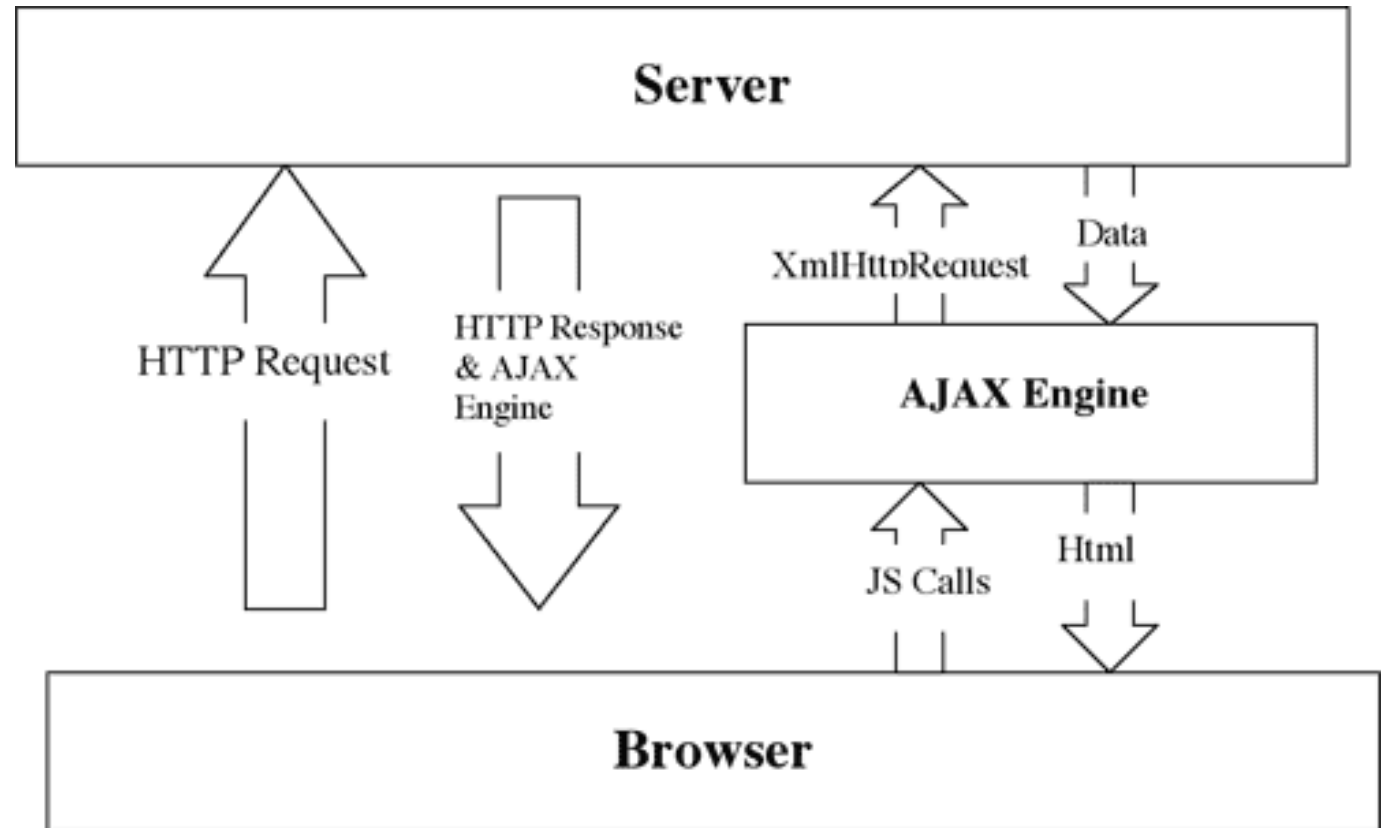


1998 - Simple Object Access Protocol (SOAP)

XMLHTTP and "AJAX"

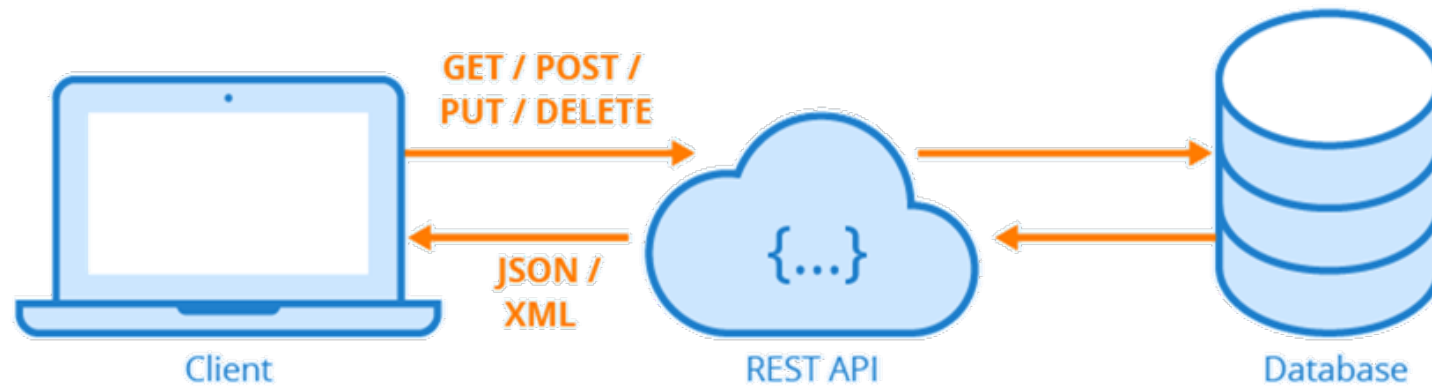
Building off the success of XML as an interchange standard, Microsoft, developed a set of mechanisms and baked them into browsers to realise a version of **Outlook for the web** which was very responsive and interactive.

This was the birth of web 2.0.



Asynchronous JavaScript and XML (AJAX) is a group of web application development technologies that allow web pages to communicate with web servers in the background.

JSON and REST

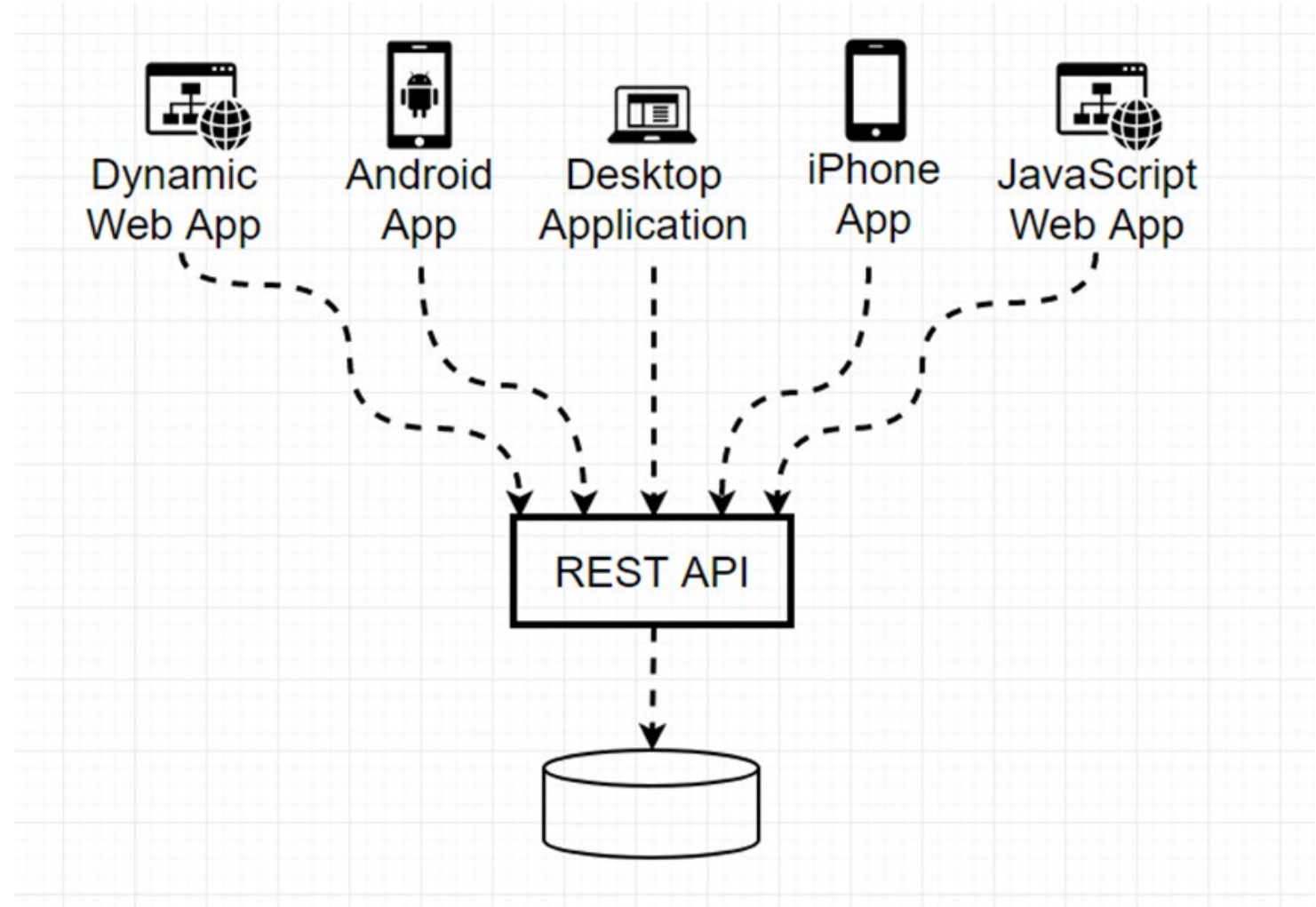


JSON

```
{
  "orderId":3,
  "productId":2,
  "quantity":4,
  "orderValue":16.60,
  "links": [
    { "rel":"product","href":"https://adventure-works.com/customers/3", "action":"GET" },
    { "rel":"product","href":"https://adventure-works.com/customers/3", "action":"PUT" }
  ]
}
```

Evolving from AJAX : **REST – 2000 , JSON – 2001**

This is a glimpse
of the modern
world



What are Cloud Native Solutions?

Here are set of tools, concepts and technologies which may be used to realise a cloud native software solution.

The CNCF define cloud native as:

“Cloud native technologies empower organizations to build and run scalable applications in modern, dynamic environments such as public, private, and hybrid clouds. Containers, service meshes, microservices, immutable infrastructure, and declarative APIs exemplify this approach.

These techniques enable loosely coupled systems that are resilient, manageable, and observable. Combined with robust automation, they allow engineers to make high-impact changes frequently and predictably with minimal toil.”



Microsoft Azure

Our Platform of Choice



Home > Education



Education | Get started



Overview

Get started

Learning resources

Roles

Software

Learning

Templates

My account

Profile

Need help?

Support

Welcome to the Azure Education Hub!

Whether you're a student getting started, an educator teaching advanced workloads, or just interested in building your cloud skills, we've got the development resources you need. [Learn more](#)



Explore Azure roles

Explore Azure roles to start building the key cloud skills you'll need to be successful in leading technology careers.

[Launch your career](#)



Redeem student credits

Start building the future with Azure for Students! Get \$100 in Azure credit when you activate your Azure for Students offer.

[Learn more](#)

[Activate offer](#)



Download free software

Gain access to full versions of professional developer tools for free to help you build code and deploy on your Azure subscription.

[Download software](#)

«

Search

Levels : All

Roles : All

55 Items

Title ↑↓	Levels ↑↓	Roles ↑↓	Last Modified ↑↓	Modules ↑↓	Duration ↑↓
Azure Fundamentals part 1: Describe core Azure services	Beginner	Administrator, Developer, Solutions Architect	2020-09-14	3	1 hr 41 min
Get started with artificial intelligence on Azure	Beginner	AI Engineer, Data Scientist, Developer	2020-07-17	1	34 min
Explore computer vision in Microsoft Azure	Beginner	AI Engineer, Data Scientist, Developer, Solutions Architect	2020-07-03	6	2 hr 56 min
Explore natural language processing	Beginner	AI Engineer, Data Scientist, Developer, Solutions Architect	2020-07-03	4	2 hr 13 min
Azure Fundamentals part 2: Describe core Azure services	Beginner	Administrator, Developer, Solutions Architect	2020-09-14	4	2 hr 40 min
Azure Fundamentals part 3: Describe core solutions	Beginner	Administrator, Business Owner, Developer, Technician	2020-09-14	6	2 hr 20 min
Azure Data Fundamentals: Explore core data concepts	Beginner	Administrator, Business User, Data Engineer	2020-06-26	5	1 hr 39 min
Azure Data Fundamentals: Explore relational data	Beginner	Administrator, Business User, Data Engineer	2020-06-29	3	1 hr 27 min
Azure Data Fundamentals: Explore non-relational data	Beginner	Administrator, Business User, Data Engineer	2020-06-29	3	2 hr 24 min
SC-900 part 1: Describe the concepts of security in Azure	Beginner	Business User	2021-07-29	2	45 min
Azure for Researchers part 1: Introduction to Cloud	Beginner	Data Scientist, higher-ed-educator, school-leaver	2021-04-28	9	5 hr 50 min
SC-900 part 2: Describe the capabilities of Microsoft Azure	Beginner	Business User	2021-06-16	4	1 hr 28 min
Create serverless applications	Beginner, Intermediate	Developer	2021-08-18	11	8 hr 37 min
SC-900 part 3: Describe the capabilities of Microsoft Azure	Beginner	Business User	2021-02-15	6	2 hr 16 min
SC-900 part 4: Describe the capabilities of Microsoft Azure	Beginner	Business User	2021-06-16	5	2 hr 27 min
Create machine learning models	Intermediate	AI Engineer, Data Scientist, Developer	2020-08-05	5	5 hr 18 min
Azure for Researchers part 2: Cloud Security and Compliance	Beginner	Data Scientist, higher-ed-educator, school-leaver	2021-04-28	4	2 hr 22 min
Build and operate machine learning solutions	Intermediate	Data Scientist	2021-05-24	14	10 hr 11 min
Get Started Querying with Transact-SQL	Beginner	Data Analyst, Data Engineer, Data Scientist, Data Architect	2021-05-07	6	5 hr 40 min
Provision and manage Azure Cognitive Services	Intermediate	AI Engineer, Data Scientist, Developer, Solutions Architect	2021-06-28	4	3 hr 5 min
Web Development for Beginners	Beginner	Other	2021-04-26	6	1 hr 49 min
Understand data science for machine learning	Beginner	AI Engineer, Data Scientist, Developer	2021-06-30	9	7 hr 25 min
Azure Static Web Apps	Beginner	Developer, Solutions Architect	2020-11-30	5	2 hr 18 min
Build JavaScript applications with Node.js	Beginner	Developer	2020-07-17	6	2 hr 51 min

started ↗ ...

Welcome to the Azure Education Hub!

Whether you're a student getting started, an educator teaching advanced workloads, or just interested in building your cloud skills, we've got the development resources you need. [Learn more](#)



Explore Azure roles

Explore Azure roles to start building the key cloud skills you'll need to be successful in leading technology careers.

[Launch your career](#)



Redeem student credits

Start building the future with Azure for Students! Get \$100 in Azure credit when you activate your Azure for Students offer.

[Learn more](#)

[Activate offer](#)



Download free software

Gain access to full versions of professional developer tools for free to help you build code and deploy on your Azure subscription.

[Download software](#)

Services See all Marketplace See all

Cost Management + Billing

Reservations

Azure Cosmos DB

Hosts

Subscriptions

Container instances

Content Moderator

Content moderators

Custom locations

Custom vision

Resources

No results were found.

Didn't find what you were looking for?

[Try searching in Activity Log](#)

[Try searching in Azure Active Directory](#)

Searching 1 of 2 subscriptions.

Get \$100 in Azure credit when you activate your Azure for Students offer.

[Launch your career](#)

CadDo Calculate

CloudMonitor Analytics Engine - Cost Optimization, Cost Mana...

Azure Cost Management

SynCloud - AI Cloud Cost Management Solution

Documentation

[Get started: Manage cloud costs - Cloud Adoption Framework ...](#)

[Plan to manage costs for Azure Data Factory - Azure Data ...](#)

[Develop a cost model - Azure Architecture Center ...](#)

[Track costs across business units, environments, or ...](#)

Resource Groups

No results were found.

[Learn more](#)

[Activate offer](#)

[Download software](#)

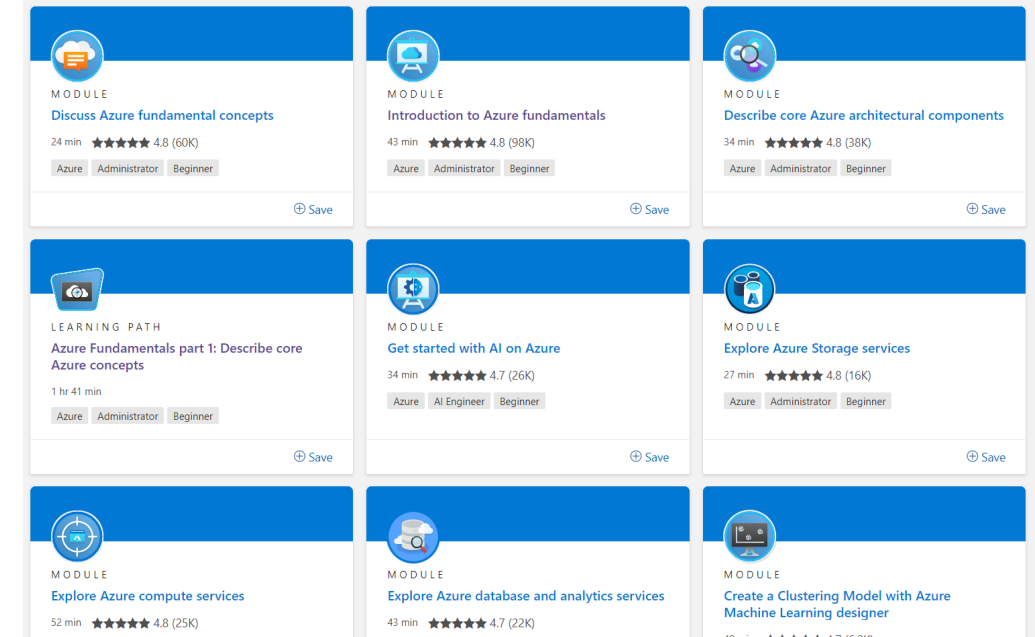
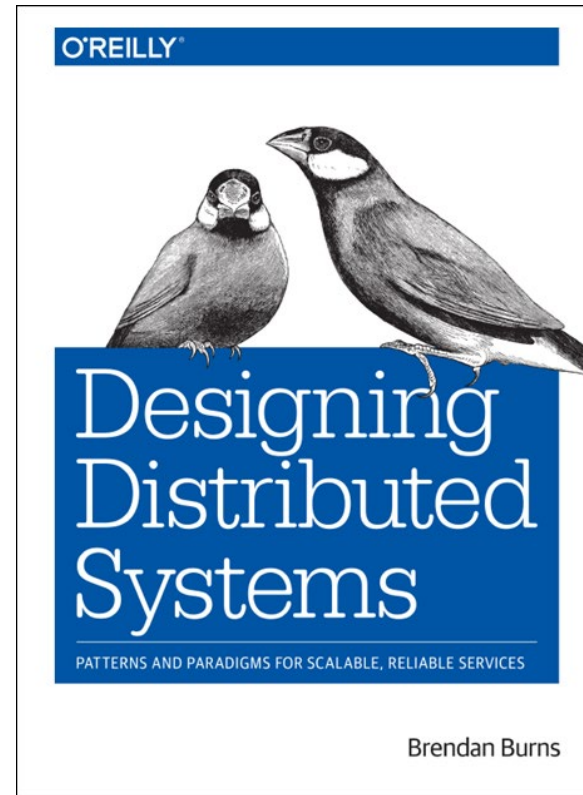
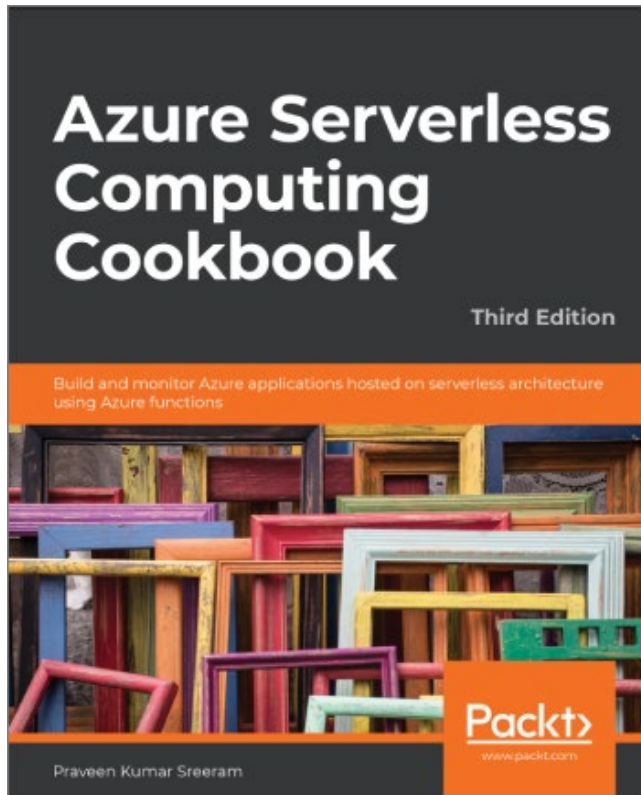


Dolla Dolla Bill, Yall.

Azure Credit

\$100 FREE





Supporting Materials



- ***Plagiarism is not tolerated***
- ***It is very easy to detect***
- ***Raw module stats can indicate plagiarism in isolation more so when other modules are mapped***
- ***Machine Learning based scanning routines, such as those in Turnitin, can easily identify plagiarised work***

Don't do it . Don't enable it. All guilty parties are penalised.



Any Questions?